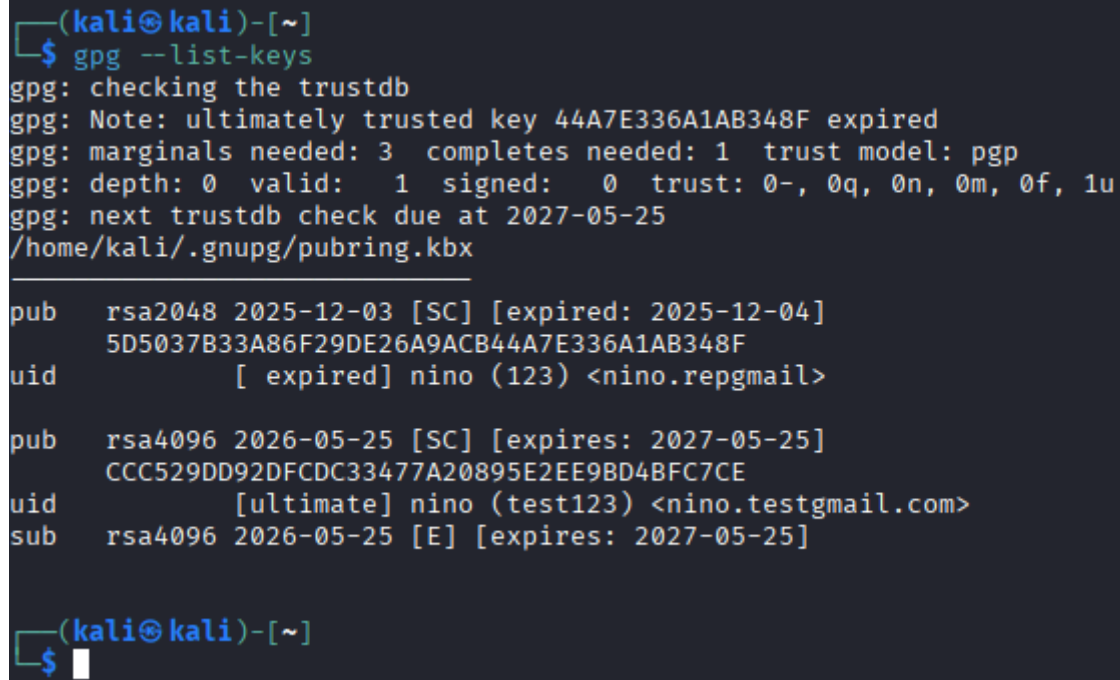


PGP Encryption and Signing

Key Generation

I generated a new GPG key pair using RSA 4096 encryption.

`gpg --full-generate-key`

A terminal window on a Kali Linux system. The prompt is (kali@kali)-[~]. The user enters 'gpg --list-keys'. The output shows GPG checking the trustdb, noting an expired key, and listing two keys: a 2048-bit RSA key for 'nino (123)' and a 4096-bit RSA key for 'nino (test123)'.

```
(kali@kali)-[~]  
$ gpg --list-keys  
gpg: checking the trustdb  
gpg: Note: ultimately trusted key 44A7E336A1AB348F expired  
gpg: marginals needed: 3 completes needed: 1 trust model: pgp  
gpg: depth: 0 valid: 1 signed: 0 trust: 0-, 0q, 0n, 0m, 0f, 1u  
gpg: next trustdb check due at 2027-05-25  
/home/kali/.gnupg/pubring.kbx  
  
pub  rsa2048 2025-12-03 [SC] [expired: 2025-12-04]  
    5D5037B33A86F29DE26A9ACB44A7E336A1AB348F  
uid          [ expired] nino (123) <nino.repgmail>  
  
pub  rsa4096 2026-05-25 [SC] [expires: 2027-05-25]  
    CCC529DD92DFCDC33477A20895E2EE9BD4BFC7CE  
uid          [ultimate] nino (test123) <nino.testgmail.com>  
sub  rsa4096 2026-05-25 [E] [expires: 2027-05-25]  
  
(kali@kali)-[~]  
$
```

Encryption and Signing

I created a text file and encrypted/signed it using GPG.

`gpg --encrypt --sign --armor --nino.testgmail.com message.txt`

The encrypted file was successfully generated.

Decryption and Verification

I decrypted the encrypted message and verified the digital signature.

`gpg --decrypt message.txt.asc`

The terminal displayed a valid signature confirmation.

```
(kali㉿kali)-[~]  
$ gpg --decrypt message.txt.asc > decrypted_message.txt  
gpg: encrypted with rsa4096 key, ID 1C36B86FBAD4647B, created 2026-05-25  
      "nino (test123) <nino.testgmail.com>"  
gpg: WARNING: server 'gpg-agent' is older than us (2.4.8 < 2.4.9)  
gpg: Note: Outdated servers may lack important security fixes.  
gpg: Note: Use the command "gpgconf --kill all" to restart them.  
gpg: Signature made Mon 25 May 2026 04:59:46 AM EDT  
gpg:                using RSA key CCC529DD92DFCDC33477A20895E2EE9BD4BFC7CE  
gpg: Good signature from "nino (test123) <nino.testgmail.com>" [ultimate]
```

Reflection

Difference between encryption and signing

Encryption protects confidentiality, while digital signing confirms authenticity and integrity.

Role of public and private key

The public key is used for encryption, while the private key is used for decryption and signing.

What happens if the encrypted file is modified?

If the encrypted or signed file is modified, the signature verification will fail and integrity cannot be confirmed.